Parity and Pell Witnesses

Smaller explicit Diophantine equations for syntactic consistency
and a reduced direct proof-calculus quartic

AI-assisted research prepared for Vladimir Reshetnikov
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Abstract

We give new fixed-coefficient Diophantine constructions whose absence of integer zeros is provably equivalent in ZFC to syntactic Con(ZFC), by composition with explicitly identified inherited calibration theorems. The main mechanism uses an odd affine coordinate to combine Pell witnesses. It excludes cancellation branches that a bare radical norm would otherwise admit. A two-root version yields 16 unrestricted integers, exact degree 112, and 119 binary arithmetic operations after a modulus-four refinement. A three-root version yields 15 unrestricted integers, degree 310, and 136 operations; a polynomial translation gives degree 306 at 139 operations. A separate sign-preserving reduction gives 20 unrestricted integers at degree 18.

We also reduce the completed direct C222 proof-calculus construction from 27,916 to 26,188 unrestricted integer variables while retaining degree four. Its complete 724,768-term polynomial is supplied. This route uses neither K215 simulation nor a Jones universal polynomial. The results are explicit improvements and tradeoffs against a pinned repository and its incoming archives, not a proof of global minimality or historical priority. The archive contains literal equations, complete coefficient bytes, mathematical proofs, independent exact checks, and reproducible builders. No new end-to-end proof-assistant certificate or exported object-language ZFC derivation is claimed.

Contents

1 What is established **3**
- 1.1 The target and the comparison . . . 3
- 1.2 The new idea and the proof boundary . . . 4

2 Fixed coefficients and the calibrated parent **4**
- 2.1 Six fixed integers . . . 4
- 2.2 Shared polynomial registers . . . 4
- 2.3 The parent and its logical interface . . . 5

3 Lemmas available before root recovery **6**

4 An odd two-root norm: sixteen integers, degree 112 **7**

5 Odd three-root recovery **8**
- 5.1 The algebraic norm . . . 8

6 The fifteen-integer polynomial **9**
6.1 A complete division-free definition 9
6.2 The unit product and the complete witness proof 9
6.3 Exact degree and operation count 10

7 A lower-degree polynomial translation **11**

8 The final modulus-four refinement **11**
8.1 Sixteen integers at 119 operations 11
8.2 Fifteen integers at 136 or 139 operations 12

9 Rescaling a sign condition: twenty integers, degree 18 **13**

10 A smaller completed direct proof-rule quartic **13**
10.1 The rank already present in the power certificate 13
10.2 Eliminating two semantic witnesses 14
10.3 Composition into one literal polynomial 15
10.4 Logical calibration and evidence boundary 15

11 Why the direct equation needs no machine **16**
11.1 Unordered Horn proof certificates 16
11.2 One product stream 16
11.3 Closing the arithmetic 16

12 Verification and practical use **17**
12.1 Exact checks and their limits 17
12.2 Reproduction 18
12.3 Research implications 18

A Compact specification of the final fifteen-variable equation **19**

B Provenance and reading order **19**

1 What is established

1.1 The target and the comparison

Fix the recursive syntactic calculus for ZFC used in ProveIt. Write $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ for its finite-refutation predicate and

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

The target is an explicit polynomial P and an ordinary mathematical proof, formalizable in ZFC, of

$$\mathrm{ZFC} \vdash (\neg\exists x \in \mathbb{Z}^n\ P(x) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

All coordinates in the small equations below range over *all integers*. A polynomial in fifteen positive integers and one in fifteen unrestricted integers make different complexity claims.

Number of unknowns, total degree, evaluator size, coefficient size, and proof complexity need not improve together. Every binary addition, subtraction, and multiplication costs one operation here, including squares and scalar multiplication. Reused registers and fixed integer leaves are free. This circuit convention is not bit complexity.

The comparison snapshot is ProveIt commit `43ac596e36c105a0e8b07c2d1f5ae200cae9c056`. Initial authenticated plugin reads used the earlier `bf2f392d73eb2ed74bd51cfdb509ecb704d04eb3`. The later snapshot includes the direct C222 round-twelve archive. The source ledger records every incoming ZIP hash. We inspected those archives as well as the collection README, which does not describe all the newer results.

Construction, all over $\mathbb{Z}$	Unknowns	Degree	Operations
Incoming saturated unit-product	16	184	135
Incoming saturated sum of squares	16	192	134
New two-root norm, mod four	**16**	**112**	**119**
New three-root norm, mod four	**15**	**310**	**136**
New centered norm, mod four	**15**	**306**	**139**
Incoming transport projection	20	20	111
New rescaled sign guard	**20**	**18**	**115**
Incoming direct C222 quartic	27,916	4	—
New direct C222 quartic	**26,188**	**4**	—

Table 1: Exact domain-matched comparisons. The sixteen-integer result improves degree and evaluator size. The other small forms are tradeoffs. The direct construction is measured by its full sparse polynomial.

The table lists the final modulus-four refinements. We first prove the slightly simpler odd-coordinate versions, at 120, 137, and 140 operations, and then save one gate in each in Section 8. All six literal circuits are retained.

The saturated-certificate archive supplies the calibrated seventeen-integer parent used below [1]. Polynomial shears and saturation/projection already supply twenty-integer degree-twenty and other reductions [2, 3]. The incoming direct construction already completes a machine-independent proof-rule route [4]; we improve it rather than claiming that architecture as new.

The repository also has equations in nine positive or ten unrestricted integer unknowns of much larger degree. General low-variable reductions, including Sun's recent ten-integer theorem, are separate baselines [8, 9]. Our fifteen-integer result is not a fewest-variable record. It supplies a much lower-degree, short, directly auditable alternative.

1.2 The new idea and the proof boundary

The main device is a witness-recovery argument: an odd total coordinate prevents precisely the cancellation cases that would destroy integrality in a three-root norm. This is more specific than invoking MRDP. The resulting objects are fully specified finite polynomials; named registers are abbreviations, not hidden unknowns.

Three kinds of evidence are distinct:

1. The new transformations and all exceptional cases are proved below.
2. Exact finite checks bind the formulas to equation files, complete fixed coefficients, operation counts, and nonzero leading terms.
3. The interpretation as syntactic ZFC refutability uses the named inherited native/compiler/calculus theorems, or the direct C222 calibration.

The third step is an ordinary mathematical dependency, not a consequence of tests or hashes. The retained status documents record unfinished end-to-end native formalization and object-language proof obligations [7]. Reusing ordinary theorems formalizable in ZFC establishes the ordinary proof of (1); it does not emit a proof file in an encoded ZFC calculus.

No assertion of Con(ZFC) is made. The equations are not claimed to have been proved unsolvable. The result is the biconditional, whose proof does not assume consistency.

2 Fixed coefficients and the calibrated parent

2.1 Six fixed integers

The small integer equations use exactly the coherent source-34 tuple:

$$\beta = 2^{1064921} - 1, \quad K_0, \quad M_C, \quad M_F, \quad A_{\mathrm{c}} = 2^{1113}, \quad B_{\mathrm{c}} = 2^{666104}.$$

Here K_0, M_C, M_F are fixed numerals, not parameters to choose later. Their complete values are supplied as minimal unsigned little-endian byte strings: $b_0, \ldots, b_{\ell-1}$ denotes $\sum_{j=0}^{\ell-1} b_j 256^j$. The binding records lengths, bit lengths, and SHA-256 values; the bytes, not the hashes, define the constants. The six raw encodings occupy 615,883 bytes. The finite recipe, literal machine table, and ordered native tile data are retained. No coefficient depends on searching for a refutation.

M_F already includes its inherited shift by β; it must not be shifted again. The compact44, compact54, and source-34 ports are not interchangeable. Every small integer equation here uses the same complete six-port binding. The inherited source establishes

$$\beta \geq 17, \quad \beta \equiv 1 \pmod 3, \quad 0 < M_C < \beta < M_F < 2\beta, \quad K_0 \geq 0, \quad A_{\mathrm{c}}, B_{\mathrm{c}} \geq 2.$$

These elementary facts suffice for the new preliminary guards. Calibration of the actual tuple remains the identified inherited theorem.

2.2 Shared polynomial registers

The following coordinates, whenever included in an equation, are independent integers:

$$J, C, z, f, h, v, w, \gamma, T, c, t, p_1, p_2, p_3.$$

Additional coordinates will be listed for each equation. Define

$$\begin{aligned}
&J_0 = 3J + 1, && q = \beta J_0 + 1, && W = 4(3w + 1), && \\
&X = qW, && Y = (2v + 1)q^3, && E = XY, && \\
&a = (X + 1)Y, && H = 4a + 3, && \Delta = a^2 + H, && (2)\\
&F = (K_0 + W)C - (q - 1)z, && \alpha = q - F - C, && && \\
&R = (q - F)(q^3 - q) - C(q^2 - 1) + (M_C + qM_F)J_0, && && && (3)\\
&k = R + 1 + hE, && \eta = c - Yk, && \zeta = k - \eta, && \\
&m = XY^2, && B_0 = mk, && S = k(2m + 1), && (4)\\
&V = c(Tf - 1) - Rf^2, && s = X + \gamma H. && && (5)
\end{aligned}$$

The quotient-compressed V is $cTf - c - Rf^2$, not $f^2(cT - R) - c$; confusing them would change the polynomial.

Two retained residuals are

$$\begin{aligned}
\mathcal{M} &= 2acs + s^2 - Hc^2 - 1, && (6)\\
d_{\mathrm{g}} &= (A_{\mathrm{c}}F)(B_{\mathrm{c}}C) - (q - A_{\mathrm{c}}F - B_{\mathrm{c}}C)^2, && \\
\mathrm{Pos} &= 1 + p_1^2 + p_2^2 + p_3(p_3 + 1), && \\
\mathcal{G} &= d_{\mathrm{g}}\eta\zeta - (t^2 - d_{\mathrm{g}} - \eta\zeta + 1)^2 - \mathrm{Pos}\,. && (7)
\end{aligned}$$

Thus $\mathcal{M} = (ac + s)^2 - \Delta c^2 - 1$. Every name in these displays is a polynomial abbreviation.

2.3 The parent and its logical interface

The inherited parent adds coordinates r, τ, ν and sets $I = (2r + 1)c^2$. Its five residuals are

$$\begin{aligned}
&\mathcal{F} = \tau^2 + (2\tau - k)B_0 - 1,\\
&\mathcal{M}, \qquad I^2 - \Delta(f^2 - 1),\\
&\mathcal{A} = (V + \nu)^2 - I^2\nu(2V + \nu) - 1,\\
&\mathcal{G}. && (8)
\end{aligned}$$

Their squared sum is its defining polynomial. Put

$$\tau_{\mathrm{old}} = \tau + B_0, \qquad y = V + \nu, \qquad A_0 = 1 + m(m + 1)k^2.$$

The exact centering identities are

$$\begin{aligned}
\mathcal{F} &= \tau_{\mathrm{old}}^2 - A_0, && (9)\\
\mathcal{A} &= I^2V^2 - (I^2 - 1)y^2 - 1. && (10)
\end{aligned}$$

The inherited theorem identifies existence of an integer zero of this literal parent with finite syntactic ZFC refutability [1]. Its proof restores the signed native/Pell source, including the quotient witnesses $V = of - c = jc - R$. The required divisibility $c \mid f^2 - 1$ is proved by the ordinary strong-rank argument before those omitted quotients are used. This ordering is essential.

Our reductions keep $\mathcal{M}$ and $\mathcal{G}$ and recover exactly the other three parent rows. Their existence equivalence is to this particular calibrated parent, not to an unspecified universal polynomial.

3 Lemmas available before root recovery

Lemma 3.1 (Inherited joint positivity)**.** *For integers x, y,*

$$x > 0,\ y > 0 \iff \exists t, p_1, p_2, p_3 \in \mathbb{Z} :\ xy - (t^2 - x - y + 1)^2 = \mathrm{Pos}(p_1, p_2, p_3).$$

Proof. The right side is positive, hence $xy > 0$. If $x = -a, y = -b$ with $a, b > 0$, then $(t^2+a+b+1)^2 > ab$, a contradiction.

Conversely put $N = x + y - 1 \geq 1$. Since $xy - N = (x-1)(y-1) \geq 0$, choose a neighboring square t^2 with $(t^2 - N)^2 < N$. If N is a square the error is zero; otherwise one of the squares adjacent to N has error at most $\lfloor\sqrt{N}\rfloor$, whose square is less than N. The resulting positive difference is a value of Pos: Legendre's three-square theorem represents $4n - 3$, which is 1 or 5 modulo 8. Exactly one square is odd. Writing it as $(2p_3 + 1)^2$ and the other two as $(2p_1)^2, (2p_2)^2$ gives the formula. □

Lemma 3.2 (Guard-only nonzero scales)**.** *If $\mathcal{G} = 0$, then*

$$|k| \geq 2, \quad c \neq 0, \quad \Delta > 1, \quad A_0 > 0, \quad S \neq 0, \quad S^2 \geq 4A_0 > A_0.$$

No first, strong, or auxiliary Pell equation is used here.

Proof. Lemma 3.1 gives $d_{\mathrm{g}} > 0$ and $\eta\zeta > 0$. Thus η, ζ have the same nonzero sign, $|k| \geq 2$, and $|\eta| < |k|$. The integer $J_0 = 3J + 1$ is nonzero; the actual β therefore gives $q \neq 0$ and $|q| \geq 16$. Also $W = 12w + 4$ has $|W| \geq 4$, so $|X| \geq 64$ and $|Y| \geq 16^3$. It follows that

$$|c| = |\eta + Yk| > |k|(|Y| - 1) > 0, \qquad |a + 2| = |(X + 1)Y + 2| > 2.$$

Hence $\Delta = (a + 2)^2 - 1 > 1$. For every integer m, $m(m + 1) \geq 0$, so $A_0 \geq 1$. Finally the exact identity

$$S^2 - 4A_0 = k^2 - 4 \tag{11}$$

gives the asserted separation. □

We retain the adjacent-coefficient descent from the saturated source and its package-69 predecessor [1]; it is reproduced here to make the new root-recovery argument self-contained.

Lemma 3.3 (Inherited adjacent-coefficient rational descent)**.** *If $Q > 1, V$ are integers and $y \in \mathbb{Q}$ satisfies $QV^2 - (Q-1)y^2 = 1$, then $y \in \mathbb{Z}$.*

Proof. Write $y = a/b$ in lowest terms with $b > 0$. Clearing denominators gives $b^2 \mid Q - 1$. Take absolute values of V, y. The norm excludes $V = 0$; at $V = 1$ it gives $y = 1$ and $b = 1$.

For $V > 1$ use the norm-preserving transformation

$$V' = (2Q - 1)V - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2QV.$$

The first coordinate is integral because $b^2 \mid Q - 1$. The reduced denominator of y' is exactly b, because $2Q - 1 \equiv 1 \pmod{b}$. The norm is preserved by direct expansion. Since $y > V$, $V' < V$. Also

$$Q^2V^2 - (Q - 1)^2y^2 = QV^2 + Q - 1 > 0,$$

so $(Q-1)y < QV$ and $V' > -V$. The transformed norm excludes $V' = 0$. Thus $0 < |V'| < V$. Repeated absolute-value descent through positive integers preserves the denominator until $|V| = 1$, where it is one. □

This is a theorem for adjacent coefficients, not for arbitrary rational points on arbitrary Pell conics. Its unbounded proof cannot be replaced by a finite rational-root search.

4 An odd two-root norm: sixteen integers, degree 112

Theorem 4.1 (Odd two-root extraction)**.** *Let $A, Q, d, L \in \mathbb{Z}$, with $d \neq 0$ and L odd. Put*

$$B = d(L^2 - 4A) + Q, \qquad N_2 = B^2 - 4dL^2Q.$$

Then $N_2 = 0$ if and only if there are an odd integer u and an integer s satisfying $du^2 = Q$, $s^2 = A$, and $L = u + 2s$.

Proof. For soundness, $L \neq 0$. Set $u = B/(2dL) \in \mathbb{Q}$. The norm gives $u^2 = Q/d$; the definition of B then gives $(L-u)^2 = 4A$. Thus $s = (L-u)/2$ is a rational square root of an integer, so is integral. Indeed, for a reduced fraction its denominator squared must divide its numerator squared. Then $u = L - 2s$ is an odd integer. Conversely, after $Q = du^2$ the identity

$$N_2 = d^2\big((L-u)^2 - 4A\big)\big((L+u)^2 - 4A\big)$$

proves the claim. □

Parity both prevents a zero extraction denominator and restores the omitted multiplier in the required odd class.

Use the sixteen coordinates

$$J, C, z, f, h, n_0, v, w, \nu, \gamma, T, c, t, p_1, p_2, p_3 \in \mathbb{Z}.$$

In addition to the shared definitions, set

$$\begin{aligned} n &= 2n_0 + 1, & d &= c^4, & Q &= \Delta(f^2 - 1), \\ L &= n + 2B_0, & G_0 &= n^2 + 4B_0(n-k) - 4, & & \\ B &= dG_0 + Q, & N_2 &= B^2 - 4dL^2Q, & & \\ \mathcal{A}_2 &= (V+\nu)^2 - Q\nu(2V+\nu) - 1. & & & & \end{aligned} \tag{12}$$

The complete equation is

$$\boxed{P_{16} = \mathcal{M}^2 + \mathcal{G}^2 + \mathcal{A}_2^2 + N_2^2 = 0.} \tag{13}$$

The identity $G_0 = (n + 2B_0)^2 - 4A_0$ cancels the largest first-Pell terms before evaluation.

Theorem 4.2. *P_{16} has an integer zero if and only if the parent (8) does. Consequently it satisfies (1).*

Proof. At a new zero each residual vanishes, and the guard gives $c \neq 0$. Apply Theorem 4.1 with $A = A_0, d = c^4$ to obtain $du^2 = Q$, $\tau_{\text{old}}^2 = A_0$, and $L = u + 2\tau_{\text{old}}$, with u odd and τ_{old} integral. Restore

$$r = (u-1)/2, \qquad \tau = \tau_{\text{old}} - B_0, \qquad I = uc^2.$$

The first and strong parent rows vanish. Since $I^2 = Q$, the retained auxiliary row is exactly the old one. The main and guard rows are unchanged.

Conversely take a parent zero and set $n_0 = r + \tau$. Then $L = (2r+1) + 2(\tau + B_0)$, so the two-root identity gives $N_2 = 0$. All other new rows vanish. Compose with the parent/refutation theorem. □

The literal division-free circuit has 120 live operations. The core norm has exact degree 56, the guard degree 44, and the other residuals smaller degree. Hence P_{16} has exact degree 112. A nonzero coefficient certificate is

$$[c^{16}J^{60}w^{12}v^{16}h^8]P_{16} = 2^{48}3^{72}\beta^{60} > 0.$$

The verifier checks the exact upper bound and this coefficient after specializing the actual tuple. Compared with the incoming 16/184/135 point, degree falls by 72 and evaluator size by 15 operations. This two-root proof does not require rational Pell descent.

5 Odd three-root recovery

5.1 The algebraic norm

For rational a, b, c, define

$$\mathscr{N}(L; a, b, c) = \big((L^2 - a - b - c)^2 - 4(ab + ac + bc)\big)^2 - 64abcL^2. \tag{14}$$

Over a field containing their square roots, this is the exact product

$$\prod_{\epsilon_1,\epsilon_2,\epsilon_3\in\{-1,1\}} (L - \epsilon_1\sqrt{a} - \epsilon_2\sqrt{b} - \epsilon_3\sqrt{c}).$$

A zero supplies a signed sum of roots, but does not by itself make the individual roots rational. The next argument removes that obstruction.

Lemma 5.1. *If a signed sum of square roots of three positive rationals is a nonzero rational, either all roots are rational, or exactly two irrational roots cancel and the remaining root is rational.*

Proof. In the finite multiquadratic extension generated by the roots, group the terms by their square class in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$. Distinct square classes have linearly independent square roots over $\mathbb{Q}$, as follows from the character basis of the multiquadratic extension. The trivial square class supplies the rational part, so at least one term is rational. If any term is irrational, the other irrational term must lie in the same class and cancel it. □

Theorem 5.2 (Odd three-root Pell merger)**.** *Let A, Q, S, V, c, L be integers such that*

$$A > 0, \quad c \neq 0, \quad S^2 > A, \quad Q \neq 1, \quad L \text{ is odd.}$$

Put $d = c^4$ and

$$a = 4A, \qquad b = Q/d, \qquad c_* = \frac{4S^2(QV^2 - 1)}{Q - 1}.$$

Then $\mathscr{N}(L; a, b, c_*) = 0$ if and only if there are integers u, τ, y, with u odd, satisfying

$$du^2 = Q, \qquad \tau^2 = A, \qquad QV^2 - (Q - 1)y^2 = 1, \qquad L = u + 2\tau + 2Sy. \tag{15}$$

Proof. Suppose the norm vanishes. Its factorization supplies some signed root sum. Establish the signs before applying a positive-radical argument.

If $Q < 0$, $a, c_* > 0$ but $b < 0$. The signed sum has exactly one nonzero purely imaginary term and two real terms, and cannot equal real L. This is impossible.

If $Q = 0$, the factorization gives $L = \pm 2\sqrt{A} \pm 2S$. Its rationality makes $\sqrt{A}$ rational, hence integral. The right side is even, contrary to odd L. The value $Q = 1$ is excluded by hypothesis; hence $Q > 1$. If $V = 0$, then $a, b > 0$ and $c_* < 0$, again giving one nonzero imaginary term. Therefore $|V| \geq 1$.

Now put $D_y = (QV^2 - 1)/(Q - 1) \geq 1$. All radicands are positive and

$$c_* = 4S^2 D_y \geq 4S^2 > 4A = a.$$

In particular the first and third radicals cannot cancel.

By Lemma 5.1, if not all roots are rational, two irrational ones must cancel. If the first and second cancel, the remaining rational root is $2Sy$ with rational y on $QV^2 - (Q - 1)y^2 = 1$. Lemma 3.3 gives $y \in \mathbb{Z}$, making L even. If the second and third cancel, the remaining root is 2τ with $\tau^2 = A$, so

$\tau \in \mathbb{Z}$ and L is even. The first and third cannot cancel by the strict inequality above. All possibilities contradict oddness. Thus all three roots are rational.

Choose the signs supplied by the vanishing norm factor. Rational-square integrality gives $\tau \in \mathbb{Z}$, and Lemma 3.3 gives $y \in \mathbb{Z}$. For either sign of nonzero S, the third signed root may be written $2Sy$. The remaining root is $u = L - 2\tau - 2Sy$, an odd integer whose square is Q/d. This proves (15).

Conversely, those equations supply one zero factor in the displayed factorization of $\mathscr{N}$. □

Remark 5.3. The strong radicand Q/d is allowed to coincide with either other radicand. Such equality is a genuine loophole for a bare norm construction. Here any cancellation involving it forces the prescribed odd coordinate to be even. Only the two even-weighted roots require strict separation.

6 The fifteen-integer polynomial

6.1 A complete division-free definition

The independent coordinates are exactly

$$\boxed{J, C, z, f, h, n_0, v, w, \gamma, T, c, t, p_1, p_2, p_3 \in \mathbb{Z}.}$$

Use the shared registers of Section 2 and define

$$\begin{aligned}
A_0 &= 1 + m(m+1)k^2, & n &= 2n_0 + 1, & d &= c^4, \\
Q &= \Delta(f^2 - 1), & \delta &= Q - 1, & D &= d\delta, \\
L &= n + 2B_0 + 2SV, & & & & \qquad (16) \\
\overline{A} &= 4A_0 D, & \overline{B} &= Q\delta, & \overline{C} &= 4S^2 d(QV^2 - 1), \\
U &= DL^2 - \overline{A} - \overline{B} - \overline{C}, & & & & \\
T_3 &= U^2 - 4(\overline{AB} + \overline{AC} + \overline{BC}), & & & & \\
N_3 &= T_3^2 - 64\overline{ABC}\, DL^2, & & & & \qquad (17) \\
G &= \mathcal{M}^2 + \mathcal{G}^2. & & & &
\end{aligned}$$

The equation is

$$\boxed{P_{15} = G + N_3 + GN_3 = 0.} \qquad (18)$$

All products and exponents are fixed. The rational expressions in Theorem 5.2 occur only in its proof; the polynomial has no division. The exact denominator-clearing identity is

$$N_3 = D^4 \mathscr{N}\left(L; 4A_0, \frac{Q}{d}, \frac{4S^2(QV^2 - 1)}{Q - 1}\right)$$

whenever $D \neq 0$.

6.2 The unit product and the complete witness proof

The finalizer saves degree:

$$P_{15} = (1 + G)(1 + N_3) - 1.$$

Since G is a sum of integer squares, $1 + G$ is a positive integer. If its product with $1 + N_3$ equals one, then $1 + G = 1$ and $N_3 = 0$. Conversely $G = N_3 = 0$ makes the equation vanish. No assumption that N_3 is nonnegative is made.

Theorem 6.1. *P_{15} has an integer zero if and only if the calibrated parent (8) does. In particular*

$$\mathrm{ZFC} \vdash \Big(\neg\exists x \in \mathbb{Z}^{15} : P_{15}(x) = 0\Big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Proof. At a new zero, $\mathcal{M} = \mathcal{G} = N_3 = 0$. The guard gives $c \neq 0, \Delta > 1, A_0 > 0, S^2 > A_0$. Also $Q = \Delta(f^2 - 1) \neq 1$, because an integer multiple of $\Delta > 1$ cannot be one. Hence $D = c^4(Q - 1) \neq 0$. The coordinate L is odd.

Apply Theorem 5.2 to the cleared norm. Recover odd u and integers τ_{old}, y, and restore

$$r = (u - 1)/2, \qquad \tau = \tau_{\mathrm{old}} - B_0, \qquad \nu = y - V.$$

With $I = uc^2$, the recovered $I^2 = Q$ is the parent strong equation. Identities (9) and (10) restore its first and auxiliary equations. The other two were retained.

Conversely, from a parent zero choose

$$n_0 = r + \tau + S\nu. \tag{19}$$

Then, with $u = 2r + 1$, $\tau_{\mathrm{old}} = \tau + B_0$ and $y = \nu + V$, we have $L = u + 2\tau_{\mathrm{old}} + 2Sy$. One norm factor vanishes, so $N_3 = 0$. The retained rows give $G = 0$ and hence $P_{15} = 0$. The inherited parent/refutation theorem gives the claimed ZFC equivalence. □

The soundness proof uses rational expressions only as proof tools and then proves the recovered values integral. The reverse map (19) is polynomial in the old integer coordinates. Existential equivalence is the relevant statement; no bijection between the complete witness sets is needed.

6.3 Exact degree and operation count

The literal circuit has 137 live binary operations. Temporarily assign weights

$$\mathrm{wdeg}(m, k, V, Q, d, n) = (10, 7, 7, 14, 4, 1).$$

The raw expression suggests degree 118 for T_3, but its leading terms cancel. Exact expansion with $S = k(2m + 1)$ and $A_0 = 1 + m(m + 1)k^2$ gives

$$\mathrm{wdeg}\, T_3 = 111, \qquad \mathrm{in}(T_3) = -512m^4k^4V^3Qd^2.$$

The other term of N_3 has degree at most 212. Consequently

$$\mathrm{wdeg}\, N_3 = 222, \qquad \mathrm{in}(N_3) = 2^{18}m^8k^8V^6Q^2d^4. \tag{20}$$

This is an exact 3,120-term polynomial computation, not a modular estimate. Substitution of the source registers retains degree 222. The highest-degree part of G has degree 88 and its product with (20) is nonzero, giving degree 310 for P_{15}. One actual coefficient certificate is

$$[c^{16}J^{184}w^{28}v^{44}h^{16}z^6f^{16}]P_{15} = 2^{118}3^{212}\beta^{184} > 0.$$

The exponents sum to 310. No genericity condition on K_0, M_C, M_F is required.

The sum-of-squares alternative $G + N_3^2 = 0$ has degree 444. The positive-unit finalizer saves 134 degrees at one additional binary operation. Its validity is the integer-unit argument above.

7 A lower-degree polynomial translation

Replace only the definition of n by

$$n = 2(n_0 - hYR) + 1. \tag{21}$$

This is an invertible polynomial change in the integer coordinate n_0: h, Y, R do not involve that coordinate. Oddness is unchanged, and the completeness map becomes $n_0 = r + \tau + S\nu + hYR$.

The motivation is the leading correction in the near-root branch: SV/Q has polynomial leading part $-2hYR$ after the source substitutions. The observation suggests a translation; exact expansion proves its effect. The certificate uses the authentic expressions

$$m = XY^2, \quad k = R + 1 + hXY, \quad V = c(Tf - 1) - Rf^2,$$

$$Q = ((X+1)^2Y^2 + 4(X+1)Y + 3)(f^2 - 1).$$

Give X, Y, R weights $2, 4, 5$, and h, f, c, T, n_0 weight one. The inner norm expands to 61,891 terms, of exact weighted degree 109, with leading form

$$512X^9Y^{13}R^3h^3f^6c^8\big[Rf^2 + hY(X - 2f^2)\big].$$

The subtracted triple-radical tail has degree at most 212. Thus the norm has degree 218, and the final polynomial has degree $88 + 218 = 306$. On the actual source substitution a certificate is

$$[J^{184}w^{26}v^{42}z^8h^{14}f^{16}c^{16}]P_{15,\mathrm{c}} = 2^{112}3^{210}\beta^{184} > 0.$$

The translated circuit has 140 operations. Evaluating $2(n_0 - hYR) + 1$ costs three more operations than $2n_0 + 1$. Both fifteen-variable equations are retained: degree 310 at 137 operations and degree 306 at 140 operations.

8 The final modulus-four refinement

The authentic source has a useful residue property without any Pell equation: $W = 4(3w + 1)$ makes X, and hence $m = XY^2$, divisible by four. Consequently

$$B_0 = mk \equiv 0 \pmod 4, \qquad A_0 = 1 + m(m+1)k^2 \equiv 1 \pmod 4.$$

Any integer square root of A_0 is therefore odd. Prescribing a sum congruent to two modulo four can replace the factors of two used in the preceding first-root encodings.

8.1 Sixteen integers at 119 operations

Replace (12) by

$$n = 4n_0 + 2, \qquad L = n + B_0, \qquad d = c^4, \qquad Q = \Delta(f^2 - 1),$$
$$G_4 = n^2 + B_0(2n - k) - 1 = L^2 - A_0,$$
$$B_4 = dG_4 + Q, \qquad N_{2,4} = B_4^2 - 4dL^2Q.$$

Keep $\mathcal{M}, \mathcal{G}, \mathcal{A}_2$ and use their squared sum with $N_{2,4}^2$. At a zero, $L \equiv 2 \pmod 4$, so $L \neq 0$. Extract $u = B_4/(2dL)$ and $\tau_{\mathrm{old}} = L - u$. The norm identity gives $du^2 = Q$ and $\tau_{\mathrm{old}}^2 = A_0$. Thus τ_{old} is integral and odd, and $u = L - \tau_{\mathrm{old}}$ is also an odd integer. The same parent restoration applies.

For completeness, fix the parent strong multiplier $u = 2r + 1$. Choose the sign of the private first root so that $\tau_{\text{old}} \equiv u \pmod 4$. This is always possible for an odd root and changes no other parent equation. Then $u + \tau_{\text{old}} \equiv 2 \pmod 4$, and

$$n_0 = (u + \tau_{\text{old}} - B_0 - 2)/4$$

is integral. It supplies the new zero. This is an existence-preserving sign choice, not a claim that every old value of the centered first-root coordinate is kept.

The circuit uses 119 operations. Its exact degree remains 112. The coefficient of $c^{16}J^{60}w^{12}v^{16}h^8$ is now $2^{40}3^{72}\beta^{60}$.

8.2 Fifteen integers at 136 or 139 operations

In the three-root definitions use

$$n = 4n_0 + 2, \qquad L = n + B_0 + 4SV,$$

and replace only the first and third scaled radicands by

$$\overline{A} = A_0 D, \qquad \overline{C} = 16S^2 d(QV^2 - 1).$$

Keep $\overline{B} = Q(Q-1)$, the same norm expression (17), and the unit finalizer $G + N_3 + GN_3$. The rational radicands are now

$$A_0, \qquad Q/d, \qquad 16S^2(QV^2 - 1)/(Q-1),$$

and $L \equiv 2 \pmod 4$.

The negative-radicand exclusions are unchanged. At $Q = 0$, the signed sum is $\pm\sqrt{A_0} \pm 4S$, which is odd if rational, and cannot equal even L. In the positive case, first/third cancellation is again excluded by separation. First/second cancellation would leave $4Sy$, divisible by four after rational Pell descent; second/third cancellation would leave the odd integer first root. Neither can equal $L \equiv 2 \pmod 4$. Thus all roots are rational. The recovered first root is odd, y is integral, and $u = L - \tau_{\text{old}} - 4Sy$ is an odd integer. The parent restoration is complete.

For completeness, again choose the sign of the private first root so $u + \tau_{\text{old}} \equiv 2 \pmod 4$, and set

$$n_0 = (u + \tau_{\text{old}} - B_0 - 2)/4 + S\nu.$$

This proves equivalence without an extra condition on a canonical computation witness.

The uncentered polynomial has 136 operations and degree 310. Replacing n by $4(n_0 - hYR) + 2$ gives 139 operations and degree 306; the completeness map simply adds hYR. The corresponding coefficient certificates are

$$[c^{16}J^{184}w^{28}v^{44}h^{16}z^6f^{16}]P_{15,4} = 2^{122}3^{212}\beta^{184},$$
$$[J^{184}w^{26}v^{42}z^8h^{14}f^{16}c^{16}]P_{15,4,\text{c}} = 2^{116}3^{210}\beta^{184}.$$

The independent centered expansion has 61,557 terms and inner degree 109. Its leading form is four times the one in Section 7; the norm degree is 218. These are the final small-equation evaluators recommended in Table 1.

9 Rescaling a sign condition: twenty integers, degree 18

The parent of this independent construction is the active centered joint22 equation [5]. A new coordinate $b = a - E$ allows the first two source rows to restore $E = Xb$ and $a = (X + 1)b$, retaining

$$b = (2v + 1)q^3.$$

Thus $b \neq 0$ from the actual $\beta > 1$, the form of q, and the odd factor. The monic transport pivot replaces (F, α) by C and $F = (K_0 + W)C - z(q - 1)$.

A naive substitution raises the joint guard's degree unnecessarily. Its old positive argument is $\eta(k - \eta)$, while a retained row gives $c - \eta = kb$. Set

$$u = b\eta, \qquad t' = u(c - \eta - u).$$

On that retained row,

$$t' = b^2\eta(k - \eta). \tag{22}$$

Since $b \neq 0$, positivity of the new and old arguments is equivalent. Use Lemma 3.1 on (d_{g}, t'). This preserves existence of private guard witnesses, not their old values: the nearest-square lemma regenerates the four private coordinates in each direction. It is not an equality of the two guard polynomials.

Use the strong row to replace $\Delta(f^2 - 1)$ by I^2 in the centered auxiliary row before eliminating the strong ordinate. The row is

$$(fo + y)^2 - I^2(c + y)(2fo + y - c) - 1, \qquad I = (2r + 1)c^2.$$

The eight residual degrees are $4, 5, 8, 8, 5, 8, 9, 8$. The unique degree-nine component is $-8r^2c^4fo(c + y)$. The full sum of squares therefore has exact degree eighteen and highest-degree form

$$64r^4c^8f^2o^2(c + y)^2.$$

This certificate is independent of all fixed ports. The circuit has 115 operations: 57 multiplications, 30 additions, and 28 subtractions. Compared with the incoming 20/20/111 equation, it trades four gates for a ten-percent degree decrease. The packet contains complete residual definitions, exact parent restorations, and the proof of every arithmetic row and (22).

10 A smaller completed direct proof-rule quartic

The incoming direct C222 construction has a particularly useful feature: its coefficients are ordinary printed integers, and the entire final polynomial is explicitly materialized. It uses a finite proof-flow certificate for twelve numerical Horn rules, followed by a CRT bounded family and finite arithmetic representations. No Turing-machine simulation or Jones universal polynomial occurs in this route. The incoming fixed-base affine construction has 16036 natural unknowns, or 27916 unrestricted integer unknowns after its nonnegativity conversion.

We reduce these counts to 14956 and 26188, respectively, without altering the twelve rules, the proof-flow encoding, or the CRT architecture. The change concerns a redundant truncated-subtraction certificate inside each of the 108 calls to the power representation.

10.1 The rank already present in the power certificate

All variables in this subsection range over $\mathbb{N} = \{0, 1, 2, \ldots\}$. Write the Pell residual as

$$\mathcal{N}(A; X, Y) = X^2 + Y^2 - A^2Y^2 - 1.$$

The active finite branch of the inherited power representation contains three facts:

$$a > 1, \qquad n \le w, \qquad \mathcal{N}(w+1; a, wz) = 0. \tag{23}$$

The first follows from the outer positive-parameter guard in the Pell representation, including when that representation uses its zero-index branch. The second and third are explicit residuals in the power branch.

Rank lemma. Conditions (23) imply

$$w \ge 1, \qquad z \ge 1, \qquad a > w \ge n.$$

Indeed, the norm equation is precisely

$$a^2 = 1 + w^3(w+2)z^2.$$

If $w = 0$ or $z = 0$, it gives $a^2 = 1$, contradicting $a > 1$. Consequently $w, z \ge 1$, and

$$a^2 = 1 + w^2\big(w(w+2)z^2\big) > w^2.$$

The claimed inequality follows by monotonicity of squaring on naturals. This is an unbounded arithmetic proof, not an inference from a finite search.

10.2 Eliminating two semantic witnesses

The original finite branch introduces a natural d and a natural slack s through the nonnegative disjunction polynomial

$$(a - d - n)^2\big((n - a - s)^2 + d^2\big) = 0. \tag{24}$$

It expresses $d = \max(a - n, 0)$. Under the rank lemma, the second factor cannot vanish, since $a > n$ makes $n - a - s \neq 0$. Thus $d = a - n$. The same branch contains the congruence, written with natural quotient witnesses q_0, q_1,

$$x + tq_0 = yd + m + tq_1.$$

Substitute $d = a - n$ and move yn to the left. The resulting equation is

$$x + yn + tq_0 = ya + m + tq_1. \tag{25}$$

Both sides are natural arithmetic expressions. The new compiler deletes d, s, and (24), and replaces the old congruence by (25). Every other byte of the original compiler is retained; the delivered delta checker verifies this.

Soundness and completeness, including inactive branches. When the finite branch is selected, its residuals form a sum of nonnegative expressions, so each vanishes. The rank lemma applies to the new branch as well as the old one. An old witness therefore restricts to a new witness. Conversely, a new finite-branch witness extends to an old one by setting $d = a - n$ and $s = 0$. The first factor in (24) is then zero, and the two congruences are the same integer-polynomial equation.

The inherited power representation separately covers exponent zero and zero base with positive exponent. Neither branch is changed. When one of these branches is selected, the finite expression is underneath a zero factor in the disjunction, so it imposes no condition. The erased witnesses can then be restored as $d = s = 0$. All deleted or changed natural arithmetic wires can be recomputed from their inputs. There is no requirement that an unselected subtraction or Pell branch vanish. This observation is essential to the two-way existential equivalence of the complete power graph.

10.3 Composition into one literal polynomial

Replacing an existential arithmetic graph by an equivalent graph preserves every enclosing binomial, factorial and bounded-CRT representation. The unchanged global construction uses 108 power graphs, 33 binomial graphs, 3 factorial graphs and 72 positive-divisor Euclidean-division graphs. All twelve local Horn branches and both global instantiations of the local row remain unchanged.

Let $\rho_1, \ldots, \rho_{11502}$ be the explicit quadratic residuals in `residuals_affine.json`, in 14956 natural variables. The natural equation is

$$P_{\mathbb{N}}(X) = \sum_{j=1}^{11502} \rho_j(X)^2 = 0.$$

The residuals are literal finite integer polynomials, supplied in sparse form, and their complete expansion is `equation_affine.json`. There are 3744 semantic inputs and 11212 retained nonlinear wires.

To obtain the unrestricted-integer equation, introduce three fresh integer variables u_i, v_i, w_i for each semantic input X_i, $0 \le i < 3744$, and set

$$P_{\mathbb{Z}} = P_{\mathbb{N}} + \sum_{i=0}^{3743} \big(X_i - u_i^2 - v_i^2 - w_i(w_i+1)\big)^2.$$

The polynomial has $14956 + 3 \cdot 3744 = 26188$ integer variables and total degree exactly four. Every natural is of the form $u^2 + v^2 + w(w+1)$ with integer u, v, w: apply the three-square theorem to $4X+1$, whose residue modulo eight is one or five, and separate the single odd square from the two even squares. Conversely, the displayed expression is nonnegative for every integer triple. All other wires are forced to be nonnegative by the unchanged topological equations and nonnegative affine forms, so they need no additional guards.

The complete integer expansion is `equation_integer.json`; it has 724768 nonzero monomials and maximum absolute coefficient 320307372009864. It contains no variable-exponent, factorial, binomial, bounded-quantifier or symbolic-coefficient operation.

10.4 Logical calibration and evidence boundary

The unchanged proof-flow certificate is equivalent to a finite derivation of the distinguished target in C222. The pinned logical interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The fixed Horn coefficients are independently matched to `Code254.lean` and `KCode222.lean`; the names of those files do not introduce a Turing-machine simulation into this construction. The inherited uniform CRT/Pell proof, the normalization proved above, and the finite circuit lifting yield ordinary mathematical proofs formalizable in ZFC of

$$\mathrm{ZFC} \vdash \big(\neg\exists X \in \mathbb{N}^{14956} : P_{\mathbb{N}}(X) = 0\big) \iff \mathrm{Con}(\mathrm{ZFC}),$$

and the corresponding statement for $P_{\mathbb{Z}}$ over $\mathbb{Z}^{26188}$. No consistency assumption is used in either direction.

The exact finite audits check all arithmetic-interface polynomials, compositions, Horn branches, source substitutions, gate residuals, nonnegative affine eliminations and full expanded polynomial identities. The original arithmetic and logical source files are supplied with immutable hashes. These checks do not amount to a newly compiled Lean or Rocq proof of the complete construction. The claimed result is an ordinary proof with fully materialized and exactly audited polynomial artifacts.

This is a strict reduction of the incoming completed direct construction. It is not a global fewest-variable result, and it does not dominate the repository's native low-variable equations. The elementary

rank argument is not claimed to be historically new; its useful application here is the removal of a redundant branch certificate from every power call.

11 Why the direct equation needs no machine

The direct component offers a complementary notion of simplicity: its coefficients are ordinary modest integers, with no native coefficient payload. Its predecessor already supplies the direct architecture; the new power normalization reduces the complete measured output.

11.1 Unordered Horn proof certificates

Each numerical C222 rule has one conclusion and at most two premises, padded by truth label 0. A finite proof tree gives a nonnegative sum of rule balance vectors equal to e_1, counting positive labels and ignoring label 0.

Conversely, close the finitely supported rules under derivation from their zero-premise instances. If target 1 lay outside the closure, sum the balance coordinates outside it. Every rule contributes at most zero: an outside conclusion needs an outside premise, otherwise closure would add it. The target contributes one, a contradiction. Single conclusions and nonnegative multiplicities are essential; arbitrary signed balances would not give this implication.

11.2 One product stream

For labels at most m and N rule instances use linear weights $1 + jX$, and put

$$A(X) = \prod_i (1 + c_i X), \qquad B(X) = (1 + X) \prod_i (1 + p_i X)(1 + q_i X).$$

Their coefficient sums are bounded by $(m + 1)^N$ and $2(m + 1)^{2N}$. Since $m + 1 \le 2^m$ for all $m \in \mathbb{N}$, choose

$$M_0 = 2^{2mN+1}, \qquad M = M_0 + 1.$$

Every coefficient of $A - B$ has absolute value less than M. Thus equality at M implies polynomial equality: a least nonzero coefficient would be divisible by M but smaller in absolute value. Multiplicities of the distinct rational roots $-1/j$ recover the balance.

The recurrence

$$S_{i+1}(1 + c_i M) = S_i(1 + p_i M)(1 + q_i M), \qquad S_0 = M + 1, \quad S_N = 1$$

encodes that equality. A proof tree supplies integral accumulator values from its open obligations. Conversely all factors are positive, so telescoping is legitimate. The label bound m is chosen before M and the accumulator. A separate CRT bound on witnesses is introduced later, preventing a circular bound.

11.3 Closing the arithmetic

The predecessor's beta-stream, bounded-CRT certificate, Pell power graph, factorial graph, and binomial graph replace all the displayed variable exponents and bounded quantifiers by finite existential polynomial certificates, with boundary cases included. The normalization changes only the redundant truncated-subtraction fragment proved above.

Retaining nonlinear wires and substituting nonnegative affine wires gives the literal quadratic residual system. Its squared sum is quartic. Only semantic inputs need three-square sign guards: all other wire signs follow by induction through their defining equations.

Complete direct construction	Incoming	New
Natural variables	16,036	14,956
Unrestricted integer variables	27,916	26,188
Degree	4	4
Semantic natural inputs	3,960	3,744
Retained nonlinear wires	12,076	11,212
Quadratic residuals	12,366	11,502
Expanded natural monomials	700,865	670,455
Expanded integer monomials	758,275	724,768
Maximum absolute integer coefficient	320,307,372,009,918	320,307,372,009,864

Table 2: A complete global reduction, not only a local witness count. The integer variable count falls by 1,728, about 6.19%.

12 Verification and practical use

12.1 Exact checks and their limits

The principal exact checks establish:

- equality of residual formulas and literal circuits, variable closure, topological order, liveness, and operation counts;
- norm identities, denominator clearings, and witness-restoration identities;
- exact upper degrees, cancellations, and nonzero highest-degree coefficients on the actual fixed tuple;
- complete coefficient byte bindings and retained source identities;
- every transformed parent arithmetic row and the sign rescaling identity of the twenty-integer equation;
- the direct compiler's single changed fragment, all 108 normalized power interfaces, all arithmetic compositions, all twelve Horn branches, global substitutions, and complete final sparse polynomial expansions.

The integer direct polynomial has 724,768 nonzero monomials. Its JSON term format is an integer coefficient and a list of zero-based variable indices; repeated indices are powers. This is an exact sparse polynomial, not a symbolic instruction to discover coefficients.

The small equations use ordered registers and declared fixed leaves. Their circuit files permit only binary addition, subtraction, and multiplication. Neither representation contains a variable-exponent operation or oracle.

Exact finite algebra does not establish consistency, unbounded Pell classification, native computation calibration, or an encoded internal-ZFC derivation. Those are separate mathematical claims, with the universal new proofs and inherited dependencies stated explicitly.

12.2 Reproduction

Use the archive README and the root verification driver. The new small-equation checks and direct replay use Python's standard library. PDFLaTeX rebuilds this report without shell escape or a network. The final manifest authenticates release files and excludes itself.

The artifact count and the finite degree certificates are reproducible. PDF bytes can change between TeX distributions without changing the mathematical source. All frozen coefficient payloads and polynomial definitions must remain identical.

12.3 Research implications

The useful general lesson is that a prescribed residue class can replace some large radical-separation bounds. The three-root argument does not automatically extend to four roots: an irrational pair can cancel while two rational terms remain, creating new residue cases. Writing a larger norm is therefore not yet a fourteen-variable construction.

The direct branch has a different bottleneck: repeated arithmetic representations, rather than the fixed logical calculus. Further local graph improvements should be judged only after composition into the complete finite equation. A local reduction in variables or degree is not by itself a global record.

No optimality lower bound or worldwide priority claim is supplied. The concrete contributions are new calibrated finite equations, explicit two-way witness proofs, and audited improvements against named inputs.

A Compact specification of the final fifteen-variable equation

This is the recommended centered modulus-four equation, of exact degree 306 and evaluator size 139. All inputs are the fifteen integers of Section 6; all other symbols are polynomial registers.

$$
\begin{aligned}
J_0 &= 3J+1, & q &= \beta J_0+1, & W &= 4(3w+1),\\
X &= qW, & Y &= (2v+1)q^3, & E &= XY,\\
a &= (X+1)Y, & H &= 4a+3, & \Delta &= a^2+H,\\
F &= (K_0+W)C-(q-1)z,\\
R &= (q-F)(q^3-q)-C(q^2-1)+(M_C+qM_F)J_0,\\
k &= R+1+hE, & \eta &= c-Yk, & \zeta &= k-\eta,\\
m &= XY^2, & B_0 &= mk, & S &= k(2m+1),\\
V &= c(Tf-1)-Rf^2, & s &= X+\gamma H,\\
\mathcal{M} &= 2acs+s^2-Hc^2-1,\\
d_{\mathrm{g}} &= A_{\mathrm{c}}B_{\mathrm{c}}FC-(q-A_{\mathrm{c}}F-B_{\mathrm{c}}C)^2,\\
\mathrm{Pos} &= 1+p_1^2+p_2^2+p_3(p_3+1),\\
\mathcal{G} &= d_{\mathrm{g}}\eta\zeta-(t^2-d_{\mathrm{g}}-\eta\zeta+1)^2-\mathrm{Pos},\\
A_0 &= 1+m(m+1)k^2, & d &= c^4, & Q &= \Delta(f^2-1),\\
n &= 4(n_0-hYR)+2, & \delta &= Q-1, & D &= d\delta,\\
L &= n+B_0+4SV,\\
\overline{A} &= A_0D, & \overline{B} &= Q\delta, & \overline{C} &= 16S^2d(QV^2-1),\\
U &= DL^2-\overline{A}-\overline{B}-\overline{C},\\
T_3 &= U^2-4(\overline{AB}+\overline{AC}+\overline{BC}),\\
N_3 &= T_3^2-64\overline{ABC}\,DL^2,\\
G &= \mathcal{M}^2+\mathcal{G}^2, & P_{15} &= G+N_3+GN_3.
\end{aligned}
$$

The uncentered degree-310 version changes only n to $4n_0+2$. The literal counted circuit shares products and uses cancellation identities; counting the typeset display by eye need not give the same optimized operation count.

B Provenance and reading order

The snapshot ledger and source notices are in `provenance/`. The small-coefficient direct equation and six-port native equations are separate components. To assess the main new result, read the root-merger theorems and parent restorations. For the final consistency interpretation, follow the retained parent and ordinary calibration proofs. For the literal finite claims, run the exact checkers.

The ordinary proofs were reviewed within this AI-assisted collaboration, not by external human referees. The package is a new research contribution; it does not modify or replace earlier repository packages.

References

[1] ProveIt incoming contribution, *Saturated Pell certificates and quotient recovery for Con(ZFC)*, integer component, Sections 3–8. https://github.com/VladimirReshetnikov/ProveIt/blob/43ac596e36c105a0e8b07c2d1f5ae200cae9c056/docs/incoming/ConZFC_Saturated_Certificates_2026-10-24.zip. Relevant proofs, equations, coefficient recipe, and licenses are retained.

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[8] Zhi-Wei Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594, 2026. https://arxiv.org/abs/2609.01594.

[9] James P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic 47 (1982), 549–571. https://www.cambridge.org/core/journals/journal-of-symbolic-logic/article/113EBF3D928DFC6011C52BECAE81A087.

[10] Mario Carneiro, *A Lean formalization of Matiyasevič's theorem*, arXiv:1802.01795, 2018. https://arxiv.org/abs/1802.01795. Background on formal Pell and exponentiation representations; the direct compiler uses the supplied pinned sources.